

Data Cleaning Process: Finding and Replacing NULL Values

Objective: Identify missing customer_id values in the accessories_sales table and replace them using SQL update statements.

Step 1: Identify NULL Values

The following query was used to locate records where customer_id is NULL.

```
-- finding the null values

select * from accessories_sales
where customer_id is null;
```

Step 2: Review the Affected Records

Three records were identified with missing customer IDs.

Data Output Messages Notifications										
Showing rows: 1 to 3 Page No: 1 of 1										
	transaction_id	customer_id	customer_name	email	purchase_date	product_id	category	price	quantity	total_amo
1	1610	[null]	Joseph Morales	adamstewart@moore.com	29/01/2025	1071	Toys	1659....	4	6637.56
2	1892	[null]	Sean Marshall	kaitlinlivingston@harris-moore....	22/08/2023	1040	Toys	2559....	-2	-5119.7
3	1931	[null]	Michelle Boyer	jeffreyruiz@gmail.com	01/12/2024	1068	Books	64.43	-5	-322.15

Step 3: Update Individual Missing Values

A direct UPDATE statement was used to assign a customer ID to Joseph Morales.

```
-- updating or replacing the null values
update accessories_sales
set customer_id = '726'
where customer_name = 'Joseph Morales';
```

transaction_id	customer_id	customer_name	email	purchase_date	product_id	category	price	quantity	total_amount
1892	[null]	Sean Marshall	kaitlinlivingston@harris-moore....	22/08/2023	1040	Toys	2559....	-2	-5119.7
1931	[null]	Michelle Boyer	jeffreyruiz@gmail.com	01/12/2024	1068	Books	64.43	-5	-322.15

Step 4: Replace Remaining NULL Values Using CASE

The CASE statement allows multiple NULL values to be updated in a single query.

```
12 -- other way to replace all the null values
13
14 update accessories_sales
15 set customer_id = case
16   when customer_name = 'Sean Marshall' then '727'
17   when customer_name = 'Michelle Boyer' then '728'
18   else customer_id
19 end
20 where customer_id is null;
21
22
```

UPDATE 2

Query returned successfully in 250 msec.

Step 5: Validate the Results

After the update operation, the remaining records were verified to ensure the NULL values had been replaced.

```
select * from accessories_sales
where customer_id is null;
```

transaction_id	customer_id	customer_name	email	purchase_date	product_id	category	price	quantity	total_amount	payment_method
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Analyst Notes

- NULL values were identified using the WHERE clause.
- Individual records can be corrected using UPDATE statements.
- CASE expressions provide an efficient method for updating multiple records.
- Validation queries should always be executed after data cleaning.